

REVIEWED LITERATURE INVENTORY

Read this page before using the original 2020-26 paper review that follows

Status: Useful literature inventory, but not the authoritative experimental roster or protocol.

Corrections from the latest meeting

- **Six datasets:** YelpChi, Amazon, T-Finance, T-Social, Elliptic and FDCCompCN. IEEE-CIS-G and DGraph-Fin are outside the current benchmark.
- **Eight working models:** CARE-GNN, FRAUDRE, PC-GNN, BWGNN, GHRN, SplitGNN, PMP and DGA-GNN.
- **Benchmark precedents:** GADBench and BOND guide experimental organisation; they are not two of the eight detectors.
- **Heterophily reference:** the screen-shared paper called "Tropical/Tropic" in the transcript is TROPICAL, not GHRN. GHRN remains a selected detector.
- **Experimental scope:** do not restore MLP, XGBoost, GCN or GraphSAGE as benchmark participants. They may appear only as background literature.

Protocol that supersedes conflicting suggestions in the original review

- First reproduce the author implementation using its original dataset, split, optimizer and hyperparameters. Label those results REPRODUCTION.
- Then run the controlled benchmark using persistent dataset-specific IDs, fixed validation/test sets, common metric code, common seeds, one controlled optimizer rule, equal tuning budget and the same Kaggle T4 hardware.
- Use TR40 = 40/20/40, TR30 = 30/20/40 plus 10% unused, TR20 = 20/20/40 plus 20% unused, and TR10 = 10/20/40 plus 30% unused. Training sets must be nested.
- Preserve Elliptic chronology: validation/test remain later in time; only the earlier labelled training portion changes.
- Do not make graph rewiring, edge attacks or adversarial robustness part of the guaranteed current phase. Those ideas are future extensions only.
- Never use the test set for threshold selection, early stopping, tuning or replacement decisions.

Heterophily convention

- **Global:** different-label eligible edges divided by all eligible labelled edges. State reverse-edge, self-loop, unlabelled-endpoint and relation-combination conventions.
- **Primary local result:** for every fraud node, benign-labelled one-hop neighbours divided by all eligible labelled one-hop neighbours; report mean, median and standard deviation.
- **Multi-relation datasets:** calculate relation-specific and combined values for YelpChi, Amazon and FDCCompCN where feasible.

Important ranking rule: Direct rankings are valid only for models evaluated on the same dataset and experimental condition. Elliptic and FDCCompCN each currently have one native selected repository path, so a second eligible model must be adapted before claiming a dataset-level leaderboard. Otherwise report a compatibility case study.

Prepared 4 September 2026. The original literature pages are preserved unchanged after this review sheet.

2020-26 Relevant Papers

Graph-Based Fraud Detection: A Reproducibility-First, Project-Aligned Literature Review

COMP8851 Major Project | Combined review of 2020-2026 literature | August 2026

Researcher correction applied

The official initiation document defines an open-source, reproducible benchmark of non-graph baselines, general GNNs and fraud-specific GNNs under common splits, preprocessing, optimisation, metrics and hardware. It does not define an LLM-enhancement project. Therefore, the earlier 2025-26 LLM papers are retained as frontier literature but are no longer Tier 1. Tiering below is based first on alignment with RQ1-RQ4 and second on reproducible author code/data compatibility.

How the papers were filtered

- Task compatibility: supervised anomalous-node classification in static attributed graphs is the guaranteed task. Edge-level, temporal-only, graph-level and proprietary-only studies are downgraded unless they provide important methodology.
- Reproducibility first: official author code, public data, licence clarity, dependency feasibility and compatibility with YelpChi/Amazon are part of the tier decision - not an afterthought.
- Research-question coverage: priority is given to fair comparison (RQ1), controlled imbalance/label scarcity (RQ2), efficiency/scalability (RQ3), and graph-characteristic explanations such as homophily and neighbourhood composition (RQ4).
- Complementarity: several papers solving the same problem are not all promoted. The highest-value, best-coded representatives are prioritised; the rest remain literature context.
- Primary-source verification: titles, years, venues and central claims were cross-checked against official venue/arXiv pages and the safe-graph/graph-fraud-detection-papers discovery list. Author-stated limitations are separated from researcher-synthesised gaps.

Project lens used for tiering

RQ	Focus	What counts as useful evidence
RQ1	Comparative performance	Same data, splits, preprocessing, optimisation budget and stopping criteria across non-graph, generic GNN and fraud-specific models.
RQ2	Imbalance & label scarcity	Controlled reduction of labelled nodes and training fraud prevalence while test distribution remains fixed.
RQ3	Efficiency & scalability	Training time, inference time, peak memory and parameters in one hardware/software environment.
RQ4	Fraud-graph characteristics	Relate stability/rankings to fraud-node homophily, degree, relation count and neighbourhood label composition.

Overall tier list

Tier	Role	Papers	Count
Tier 1	Foundation / direct target	GADBench; PC-GNN; CARE-GNN; BWGNN; PMP; GAD in the Wild; MonTi (optional novelty)	7
Tier 2	Significant comparators / experiment design	GHRN; SplitGNN; DGA-GNN; DiG-In-GNN; SEC-GFD; Barely Supervised; FraudGT; GGA; BOND; DGraph; CGNN; MimbFD; DCL-GFD; TravelFraudBench; GraphConsis; GAGA; LEX-GNN; GAAP	18
Tier 3	Literature context / future directions	Selected older alternatives, deployment/explainability papers, and recent LLM/temporal frontier work that does not match the guaranteed build.	24

Reproducibility legend: HIGH = official code and practical public-data path; MEDIUM = code/data exists but adaptation or domain mismatch is material; LOW = no clear author code or proprietary/irreproducible dependency. “Perfect fit” refers to the official RQs, not paper quality.

Reproduction toolbox note - DGFraud / DGFraud-TF2: These are not counted as papers, but they are useful engineering references because they package multiple graph-fraud baselines and public fraud datasets. Use them to accelerate reproduction checks and compare implementation details; keep the project’s own split registry, evaluator, configs and adaptation log as the final source of truth.

Tier 1 - Foundation Papers and Direct Research-Gap Targets

These are the papers to take into the supervisor meeting. Each either defines the benchmark itself, supplies a highly reproducible core fraud model, or exposes a concrete limitation/future-work slice that the project can feasibly test without changing its central task.

1. GADBench: Revisiting and Benchmarking Supervised Graph Anomaly Detection

2023 | NeurIPS Datasets and Benchmarks | Benchmarking & reproducibility

Project alignment: PERFECT FIT

Reproducibility: HIGH

Problem Statement Published graph-anomaly and fraud models are difficult to compare because papers use different datasets, preprocessing, splits, tuning budgets, metrics and hardware. It is also unclear whether specialised GNNs really outperform strong tree ensembles, or whether their cost is justified at larger scale.

Simplified Abstract GADBench builds a common supervised anomalous-node benchmark covering 29 models and ten real-world datasets. It compares classical methods, tree ensembles, general GNNs and anomaly-specific GNNs under one framework and finds that simple neighbourhood-aggregated tree ensembles can outperform specialised GNNs in some settings. The framework is released as code.

Core Contribution The contribution is not another detector. It is a controlled comparison infrastructure that makes model rankings meaningful by removing avoidable protocol differences.

Fraud Relevance This is the methodological foundation for a fraud benchmark. Fraud datasets are heavily imbalanced and graph structure varies sharply, so a leaderboard built from incompatible published numbers is especially misleading.

Project Alignment Directly matches RQ1 and RQ3, and supplies the benchmark philosophy for all four RQs. Your project narrows GADBench from broad graph anomaly detection to fraud-specific methods on shared fraud datasets, with explicit label-scarcity, imbalance and graph-characteristic stress tests.

Research Gap -> Project Objectives PERFECT MATCH. GADBench is broad rather than fraud-focused and does not fully isolate controlled training-label scarcity, training-set imbalance, fraud-node homophily and neighbourhood label composition as explanatory variables. Your project can keep its fair-comparison machinery while making those fraud-specific axes first-class.

Limitations & Future Work Researcher synthesis from the paper scope: GADBench establishes a broad supervised benchmark but leaves room for a fraud-focused benchmark that connects model rankings to fraud-specific graph structure and controlled supervision regimes. This is the central gap stated in your initiation document.

Exact Limitation / Future-Work Slice to Target Target a tractable extension rather than reproducing all 29 models: reuse the benchmark discipline, then add persistent splits, controlled label budgets, controlled class ratios, fraud-homophily statistics and common runtime/memory profiling for a smaller, representative fraud model suite.

Reproducibility & Code Official repository: squareRoot3/GADBench. Public benchmark code and dataset preparation are available. Integration risk: LOW-MEDIUM because the project can reuse concepts/components without reproducing the full benchmark.

Experiment Design Value Use one split registry and evaluator for MLP/XGBoost+neighbour features, GCN, GraphSAGE and selected fraud-specific models. Report mean +/- SD across seeds; PR-AUC primary; runtime, inference, peak memory and parameter count under one environment.

2. Pick and Choose: A GNN-based Imbalanced Learning Approach for Fraud Detection (PC-GNN)

2021 | The Web Conference | Class imbalance & neighbour sampling

Project alignment: PERFECT FIT

Reproducibility: HIGH

Problem Statement Fraud nodes are rare, and a fraudulent node can be surrounded by many benign neighbours. Ordinary mini-batch sampling and aggregation therefore expose the model to much more majority-class information than fraud information.

Simplified Abstract PC-GNN changes both training-node sampling and neighbourhood sampling. Label-balanced sampling prevents the training batch from being dominated by benign nodes, while a learned neighbour-selection mechanism favours informative neighbours before aggregation. The method was designed specifically for graph fraud detection under imbalance.

Core Contribution Treats class imbalance as a graph-learning and sampling problem, not only a loss-reweighting problem.

Fraud Relevance Extreme imbalance is unavoidable in fraud. PC-GNN also captures the fact that every neighbour of a fraudster is not equally useful, which is important when attackers connect to legitimate users for camouflage.

Project Alignment One of the clearest Tier-1 implementation candidates. It maps directly to RQ2 and supports RQ4 through neighbour-composition analysis. It is already named as a target specialised method in the project initiation.

Research Gap -> Project Objectives PERFECT MATCH. The original paper demonstrates a specialised sampling strategy but does not tell us how its advantage changes under a controlled series of fraud ratios, label budgets and neighbourhood compositions when all competing models receive the same split and tuning budget.

Limitations & Future Work AUTHOR-STATED future direction: beyond label-aware graph sampling, learning a new graph structure for the imbalanced graph is identified as promising future work. The paper also predates newer message-imbalance and heterophily methods.

Exact Limitation / Future-Work Slice to Target A feasible partial future-work contribution is not to invent a full graph-structure learner, but to test PC-GNN under controlled structural conditions: reduce fraud prevalence, vary labelled positives, and stratify/perturb benign-versus-fraud neighbour ratios. This directly tests when its sampling is actually needed.

Reproducibility & Code Official repository: PonderLY/PC-GNN; YelpChi configuration is provided. Integration risk: MEDIUM because the code uses an older deep-graph stack and may need adaptation to the common data interface.

Experiment Design Value Fraud-ratio sweep (for example original, 50%, 25% and 10% of available fraud training nodes), label-budget sweep, fixed test set, and neighbourhood-composition bins. Compare PC-GNN with weighted-loss GraphSAGE and PMP to separate sampling effects from message-passing effects.

3. Enhancing Graph Neural Network-based Fraud Detectors against Camouflaged Fraudsters (CARE-GNN)

2020 | CIKM | Camouflage & multi-relation learning

Project alignment: PERFECT FIT

Reproducibility: HIGH

Problem Statement Fraudsters deliberately imitate benign feature patterns and form benign-looking relations. Standard GNN aggregation can then pull fraudulent representations toward the majority class instead of exposing the fraud signal.

Simplified Abstract CARE-GNN formalises feature camouflage and relation camouflage, then uses label-aware neighbour similarity, adaptive neighbour selection and relation-level aggregation to suppress misleading relations while preserving informative ones in multi-relation fraud graphs.

Core Contribution A foundational camouflage-resistant fraud GNN and the clearest early example of designing message aggregation around adversarial fraud behaviour.

Fraud Relevance Camouflage is one of the main reasons ordinary homophily-biased GNNs can fail on YelpChi/Amazon-style graphs. CARE-GNN gives an explicit mechanism to test against newer heterophily approaches.

Project Alignment Strong RQ4 foundation and a useful historical baseline if implementation capacity permits. It uses the same family of public fraud graphs as the project and has official code.

Research Gap -> Project Objectives STRONG/PERFECT MATCH. CARE-GNN reports effectiveness under its own protocol; it does not establish how its gains compare with PC-GNN, BWGNN, PMP and newer methods under one persistent split registry, identical evaluation code and matched compute budget.

Limitations & Future Work No dedicated author future-work section was located in the verified source. Researcher-synthesised limitation: the method uses learned neighbour filtering/relations but its relative benefit as graph homophily, degree and class composition change is not isolated under a unified benchmark.

Exact Limitation / Future-Work Slice to Target Use CARE-GNN as a controlled camouflage baseline, then relate its relative improvement to measured fraud-node homophily and benign-neighbour share. This answers a mechanistic question rather than only repeating its headline accuracy.

Reproducibility & Code Official repository: YingtongDou/CARE-GNN. Public code and fraud graph data utilities are available. Integration risk: MEDIUM because dependencies are older and the original training pipeline is bespoke.

Experiment Design Value Compute per-dataset and per-fraud-node homophily; evaluate model deltas versus GraphSAGE/GCN in low-, medium- and high-heterophily subsets. Keep train/validation/test node IDs fixed across all models.

4. Rethinking Graph Neural Networks for Anomaly Detection (BWGNN)

2022 | ICML | Spectral filtering & heterophily

Project alignment: PERFECT FIT

Reproducibility: HIGH

Problem Statement Low-pass GNNs can smooth away anomaly signals. The paper asks whether anomalies have a distinctive graph-frequency signature that calls for a different spectral filter.

Simplified Abstract BWGNN identifies a spectral right-shift: stronger anomalies move more signal toward higher graph frequencies. It introduces Beta-wavelet band-pass filters that capture multiple frequency bands while remaining spatially local. It is tested on Amazon, YelpChi, T-Finance and T-Social in supervised and low-label settings.

Core Contribution Reframes graph anomaly/fraud detection as a spectral filtering problem and provides a simple, scalable alternative to complex neighbour-selection pipelines.

Fraud Relevance Fraud graphs are often heterophilic; high-frequency information can preserve differences between fraudsters and their benign neighbours that a low-pass GCN would erase.

Project Alignment Directly relevant to RQ2, RQ3 and RQ4. The exact core datasets overlap with the project, and the official implementation has already been integrated into GADBench.

Research Gap -> Project Objectives PERFECT MATCH. The project can test whether BWGNN remains strong once label scarcity, class imbalance, persistent splits and common resource profiling are controlled, and whether its advantage tracks homophily or neighbourhood composition.

Limitations & Future Work AUTHOR-STATED limitation/future work: the paper assumes all edges are trusted and explicitly leaves structural anomalies for future work.

Exact Limitation / Future-Work Slice to Target A feasible partial future-work experiment is a small structural stress test: add/remove a controlled fraction of benign-facing edges around fraud nodes, or use an existing attack generator on a subset. The core project can still remain static node classification.

Reproducibility & Code Official repository: squareRoot3/Rethinking-Anomaly-Detection; code/data released and BWGNN is integrated into GADBench. Integration risk: LOW-MEDIUM.

Experiment Design Value Benchmark BWGNN (homogeneous and multi-relation treatment where feasible) under original and reduced-label regimes. Add graph statistics and a small edge-perturbation sensitivity experiment if time allows.

5. Partitioning Message Passing for Graph Fraud Detection (PMP)

2024 | ICLR | Heterophily-homophily mixture & imbalance

Project alignment: PERFECT FIT	Reproducibility: HIGH
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Problem Statement Many fraud GNNs try to remove heterophilic neighbours, but fraud graphs contain useful information in both homophilic and heterophilic connections. Class imbalance can additionally let benign messages dominate gradients.

Simplified Abstract PMP argues that the key is not to exclude heterophilic neighbours but to distinguish them. During aggregation, neighbours with different classes are handled by distinct node-specific aggregation functions, giving each centre node an adaptive way to use both same- and different-class evidence. The paper links this spatial design to adaptive spectral filtering.

Core Contribution A message-passing paradigm expressly designed for mixed homophily/heterophily rather than graph rewiring or blanket neighbour removal.

Fraud Relevance It targets exactly the situation where fraudsters hide among normal users while still retaining some fraud-to-fraud signal.

Project Alignment Named as a target specialised method in the initiation draft. It is arguably the best Tier-1 model for RQ4 because the project already proposes measuring fraud-node homophily and neighbourhood label composition.

Research Gap -> Project Objectives PERFECT MATCH. PMP shows that partitioned aggregation works, but the project can ask the more diagnostic question: how much does it help at different measured homophily and training-label/imbalance levels under exactly the same evaluation environment?

Limitations & Future Work No explicit future-work section was found in the verified conference record. Researcher-synthesised limitation from scope: the paper establishes a new aggregation mechanism but does not provide a unified cross-family benchmark tying gains to controlled graph-characteristic strata and common resource budgets.

Exact Limitation / Future-Work Slice to Target Turn PMP into an explanatory experiment: test whether its relative gain increases as fraud-node homophily falls or benign-neighbour dominance rises. This directly operationalises the project RQ instead of merely reproducing the paper.

Reproducibility & Code Official repository: Xtra-Computing/PMP. Public implementation exists. Integration risk: MEDIUM; model logic is available but must be adapted to the project split registry and evaluator.

Experiment Design Value Homophily-stratified evaluation plus label-budget and training-imbalance sweeps. Compare to PC-GNN (sampling), BWGNN/GHRN (spectral) and GraphSAGE (generic message passing).

6. GAD in the Wild: Benchmarking Graph Anomaly Detection under Realistic Deployment Challenges

2026 | arXiv preprint | Realistic benchmark stress testing

Project alignment: PERFECT FIT	Reproducibility: HIGH
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Problem Statement Academic GAD benchmarks often use curated graphs, moderate anomaly ratios and complete attributes, so laboratory success may not survive realistic scale, extreme rarity or missing data.

Simplified Abstract The paper creates controlled benchmark variants around three deployment challenges: million-scale graphs, extreme anomaly scarcity and missing node attributes. Across nine models it finds memory failures at scale, sharp recall collapse near 0.1% anomaly prevalence and strong sensitivity to imputation for reconstruction-based methods.

Core Contribution A 2026 benchmark methodology that deliberately perturbs data conditions rather than treating the dataset as fixed.

Fraud Relevance Extreme rarity and imperfect data are normal in operational fraud systems. Its findings support the project choice to evaluate stress conditions, not only clean benchmark accuracy.

Project Alignment This is the closest recent benchmark-methodology paper to the official project RQs. It strongly validates controlled label/imbalance stress testing and resource profiling.

Research Gap -> Project Objectives PERFECT MATCH. Your project can adapt the same diagnostic philosophy to a smaller, fraud-specific static node-classification benchmark and add graph-structure variables (homophily, degree, neighbourhood class composition) that GAD in the Wild does not make central.

Limitations & Future Work AUTHOR-STATED limitations: challenge axes are evaluated separately even though real deployments combine them; hardware experiments are laboratory-scale rather than industrial-cluster studies. AUTHOR-STATED future work includes hardware efficiency on million-node graphs, even rarer anomalies (around 0.02%), and methods specialised for missing information.

Exact Limitation / Future-Work Slice to Target Solve a feasible slice: combine two stress axes that your project already owns, such as limited labels + training imbalance, and analyse them jointly with fraud homophily. Do not attempt billion-node infrastructure or every missing-feature scenario.

Reproducibility & Code Benchmark code is released via an open anonymous repository linked by the paper. Integration risk: MEDIUM because it is a new preprint, but its experimental design can be adopted even without importing the full codebase.

Experiment Design Value Use a factorial grid over label availability and fraud prevalence; report interaction effects and failure thresholds, not only one-dimensional curves. Keep test distribution fixed as required by your initiation draft.

7. MonTi: Unveiling the Threat of Fraud Gangs to Graph Neural Networks - Multi-Target Graph Injection Attacks

2025 | AAAI | Adversarial robustness

Project alignment: PERFECT FIT - OPTIONAL NOVELTY	Reproducibility: HIGH
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Problem Statement Most fraud GNNs are tested on fixed graphs, yet organised fraud gangs can inject new accounts and edges to make several target fraudsters appear benign at once.

Simplified Abstract MonTi is an attack framework, not a detector. It jointly generates injected-node attributes and edges with a transformer-based multi-target objective, allocating attack budget across several targets to simulate coordinated fraud-gang evasion. Official code and data are released.

Core Contribution Turns the informal idea of camouflage into an explicit, reproducible adversarial stress test.

Fraud Relevance Fraud is adaptive. A detector that ranks first on a clean test graph can still be fragile once attackers manipulate structure.

Project Alignment Not required by the current RQs, so it should be treated as a supervisor-approved optional extension. If included, it gives the benchmark a genuinely novel robustness layer instead of another model.

Research Gap -> Project Objectives STRONG MATCH if the supervisor wants robustness. None of the core benchmark papers establishes how the same set of fraud-specific models degrades under coordinated injection using identical splits and resource conditions.

Limitations & Future Work AUTHOR-STATED future work includes extending attacks to multi-relational graphs, inductive settings with unseen nodes/edges, theoretical generalisation analysis and stronger defences.

Exact Limitation / Future-Work Slice to Target The most realistic half-step is to adapt a simplified MonTi attack to one multi-relational core dataset (YelpChi or Amazon) and 3-4 representative models. That partially targets the authors multi-relational future direction without turning the whole project into an adversarial-learning thesis.

Reproducibility & Code Official repository: bdi-lab/MonTi, with code and data. Integration risk: HIGH relative to the core benchmark because attack pipelines add substantial complexity.

Experiment Design Value Optional robustness leaderboard: clean PR-AUC versus attacked PR-AUC, absolute/relative drop, attack success rate and extra runtime. Restrict to a representative model subset.

Tier 2 - Significant Comparators, Ablations and Experiment Designs

Tier 2 papers are important because they change how the experiment should be designed. They should not all be implemented. The early feasibility gate should promote only the subset that adds a new mechanism or stress condition not already represented in Tier 1.

1. Addressing Heterophily in Graph Anomaly Detection: A Perspective of Graph Spectrum (GHRN)

2023 | The Web Conference | Heterophily & spectral robustness | Reproducibility: HIGH

Problem Statement / Simplified Abstract Strong spectral comparator with official code. It designs a graph-heterophily-resistant network from a spectral perspective and is useful for testing whether spectral robustness explains gains on low-homophily fraud graphs.

Core & Fraud Relevance Use as a comparator to BWGNN/PMP and stratify results by fraud-node homophily. Official code: blacksingular/GHRN. DOI verified through ACM.

Project Fit / Research Gap RQ4, secondary RQ3. Experiment value: compare spectral families under common splits and measure whether rankings change with heterophily.

Limitations & Future Work No author-stated future-work claim is treated as established here unless explicitly verified in the primary source. For Tier 2, the project value is primarily experimental: use the paper to define a comparator, ablation or stress condition under the common protocol.

Reproducibility Decision HIGH priority relative to Tier 1. Promote to implementation only after the early feasibility gate confirms repository health, dataset access, framework compatibility, licence and memory requirements.

2. SplitGNN: Spectral Graph Neural Network for Fraud Detection against Heterophily

2023 | CIKM | Heterophily & spectral decomposition | Reproducibility: HIGH

Problem Statement / Simplified Abstract Fraud-specific spectral model explicitly designed for heterophily. It provides another way to separate useful frequency information and is directly reproducible.

Core & Fraud Relevance Official PyTorch/DGL implementation: Split-GNN/SplitGNN. Good Tier-2 alternative if BWGNN/PMP integration succeeds early.

Project Fit / Research Gap RQ4. Experiment value: ablate whether extra spectral complexity beats BWGNN once tuning budget and hardware are matched.

Limitations & Future Work No author-stated future-work claim is treated as established here unless explicitly verified in the primary source. For Tier 2, the project value is primarily experimental: use the paper to define a comparator, ablation or stress condition under the common protocol.

Reproducibility Decision HIGH priority relative to Tier 1. Promote to implementation only after the early feasibility gate confirms repository health, dataset access, framework compatibility, licence and memory requirements.

3. DGA-GNN: Dynamic Grouping Aggregation GNN for Fraud Detection

2024 | AAAI | Grouped messages & feature non-additivity | Reproducibility: HIGH

Problem Statement / Simplified Abstract Targets two fraud-graph properties: non-additive attributes and distinguishable neighbour groups. Decision-tree binning and feedback-driven grouping make it a useful modern architecture with official code.

Core & Fraud Relevance Official repository supports YelpChi plus Elliptic, T-Finance and T-Social. Good feasibility candidate after Tier-1 models.

Project Fit / Research Gap RQ1/RQ3/RQ4. Experiment value: test whether added grouping complexity pays off on YelpChi and under common runtime/memory profiling.

Limitations & Future Work No author-stated future-work claim is treated as established here unless explicitly verified in the primary source. For Tier 2, the project value is primarily experimental: use the paper to define a comparator, ablation or stress condition under the common protocol.

Reproducibility Decision HIGH priority relative to Tier 1. Promote to implementation only after the early feasibility gate confirms repository health, dataset access, framework compatibility, licence and memory requirements.

4. DiG-In-GNN: Discriminative Feature Guided GNN-Based Fraud Detector against Inconsistencies in Multi-Relation Fraud Graph

2024 | AAAI | Feature/structure inconsistency | Reproducibility: HIGH

Problem Statement / Simplified Abstract Uses multi-scale contrastive learning and discriminative guidance to deal with inconsistency in multi-relation fraud graphs. It is strongly aligned to YelpChi/Amazon-style relational camouflage.

Core & Fraud Relevance Official repository: GraphBerry/DiG-In-GNN. Good Tier-2 fraud-specific comparator; integration is more complex than PC-GNN/PMP.

Project Fit / Research Gap RQ4 and RQ1. Experiment value: compare inconsistency-aware guidance with neighbour sampling and partitioned message passing.

Limitations & Future Work No author-stated future-work claim is treated as established here unless explicitly verified in the primary source. For Tier 2, the project value is primarily experimental: use the paper to define a comparator, ablation or stress condition under the common protocol.

Reproducibility Decision HIGH priority relative to Tier 1. Promote to implementation only after the early feasibility gate confirms repository health, dataset access, framework compatibility, licence and memory requirements.

5. Revisiting Graph-Based Fraud Detection in Sight of Heterophily and Spectrum (SEC-GFD)

2024 | AAAI | Heterophily, spectrum & label use | Reproducibility: HIGH

Problem Statement / Simplified Abstract Combines a hybrid spectral filter with a local environmental constraint to address heterophily and under-used labels. It is a direct bridge between spectral filtering and label-scarcity questions.

Core & Fraud Relevance Official repository: Sunxkissed/SEC-GFD; four fraud datasets. Strong reserve implementation if one Tier-1 model fails feasibility.

Project Fit / Research Gap RQ2/RQ4. Experiment value: compare its label-use mechanism with PMP and PC-GNN under decreasing labels.

Limitations & Future Work No author-stated future-work claim is treated as established here unless explicitly verified in the primary source. For Tier 2, the project value is primarily experimental: use the paper to define a comparator, ablation or stress condition under the common protocol.

Reproducibility Decision HIGH priority relative to Tier 1. Promote to implementation only after the early feasibility gate confirms repository health, dataset access, framework compatibility, licence and memory requirements.

6. Barely Supervised Learning for Graph-Based Fraud Detection

2024 | AAAI | Extremely limited labels | Reproducibility: LOW-MEDIUM

Problem Statement / Simplified Abstract Uses strong/weak augmentation and consistency regularisation with edge-based disentangled representations to learn when labelled fraud samples are scarce. The research question matches the project exactly, but public author code is not clearly available in the major curated lists.

Core & Fraud Relevance Excellent experiment-design paper; weaker implementation candidate because reproducibility risk is higher.

Project Fit / Research Gap RQ2. Experiment value: define 10%, 5%, 1% label regimes and compare supervised fraud-specific models without necessarily reproducing this architecture.

Limitations & Future Work No author-stated future-work claim is treated as established here unless explicitly verified in the primary source. For Tier 2, the project value is primarily experimental: use the paper to define a comparator, ablation or stress condition under the common protocol.

Reproducibility Decision MEDIUM priority relative to Tier 1. Promote to implementation only after the early feasibility gate confirms repository health, dataset access, framework compatibility, licence and memory requirements.

7. FraudGT: A Simple, Effective, and Efficient Graph Transformer for Financial Fraud Detection

2024 | ICAIF | Efficiency & graph transformers | Reproducibility: HIGH

Problem Statement / Simplified Abstract Designed to balance effectiveness and efficiency in financial fraud detection, making it particularly relevant to the projects performance-cost analysis.

Core & Fraud Relevance Official repository: junhongmit/FraudGT. Domain/data construction differs from guaranteed core datasets, so treat as conditional extension rather than core.

Project Fit / Research Gap RQ3. Experiment value: transformer-vs-message-passing cost comparison if integration budget remains.

Limitations & Future Work No author-stated future-work claim is treated as established here unless explicitly verified in the primary source. For Tier 2, the project value is primarily experimental: use the paper to define a comparator, ablation or stress condition under the common protocol.

Reproducibility Decision HIGH priority relative to Tier 1. Promote to implementation only after the early feasibility gate confirms repository health, dataset access, framework compatibility, licence and memory requirements.

8. Generative Graph Augmentation for Minority Class in Fraud Detection

2023 | CIKM | Minority-class augmentation | Reproducibility: MEDIUM

Problem Statement / Simplified Abstract Synthesises minority-class graph evidence instead of changing the loss or neighbour sampler. This gives a conceptually different response to imbalance and is already cited in the initiation draft.

Core & Fraud Relevance Code is reported in graph-anomaly reading lists, but the pipeline is less plug-and-play than PC-GNN. Treat mainly as experiment/ablation inspiration.

Project Fit / Research Gap RQ2. Experiment value: compare graph-native sampling (PC-GNN) with augmentation-based minority support.

Limitations & Future Work No author-stated future-work claim is treated as established here unless explicitly verified in the primary source. For Tier 2, the project value is primarily experimental: use the paper to define a comparator, ablation or stress condition under the common protocol.

Reproducibility Decision MEDIUM priority relative to Tier 1. Promote to implementation only after the early feasibility gate confirms repository health, dataset access, framework compatibility, licence and memory requirements.

9. BOND: Benchmarking Unsupervised Outlier Node Detection on Static Attributed Graphs

2022 | NeurIPS | Benchmark design | Reproducibility: HIGH

Problem Statement / Simplified Abstract A key benchmarking predecessor that standardises unsupervised node-anomaly evaluation. Although the official project is supervised, its benchmark discipline and reproducibility lessons remain valuable.

Core & Fraud Relevance Use for benchmark design and reporting structure, not as a core detector. Public benchmark code is available.

Project Fit / Research Gap Methodology support. Experiment value: reproducibility reporting, common data treatment and benchmark hygiene.

Limitations & Future Work No author-stated future-work claim is treated as established here unless explicitly verified in the primary source. For Tier 2, the project value is primarily experimental: use the paper to define a comparator, ablation or stress condition under the common protocol.

Reproducibility Decision HIGH priority relative to Tier 1. Promote to implementation only after the early feasibility gate confirms repository health, dataset access, framework compatibility, licence and memory requirements.

10. DGraph: A Large-Scale Financial Dataset for Graph Anomaly Detection

2022 | NeurIPS | Large-scale financial benchmark data | Reproducibility: HIGH

Problem Statement / Simplified Abstract Provides a large public financial graph for scaling experiments, giving a realistic counterpoint to YelpChi/Amazon.

Core & Fraud Relevance Useful only as a conditional scalability dataset because current methodology guarantees static attributed YelpChi/Amazon and aims to keep the task comparable.

Project Fit / Research Gap RQ3. Experiment value: one or two scalable models only; do not force every specialised model onto DGraph.

Limitations & Future Work No author-stated future-work claim is treated as established here unless explicitly verified in the primary source. For Tier 2, the project value is primarily experimental: use the paper to define a comparator, ablation or stress condition under the common protocol.

Reproducibility Decision HIGH priority relative to Tier 1. Promote to implementation only after the early feasibility gate confirms repository health, dataset access, framework compatibility, licence and memory requirements.

11. Context-aware Graph Neural Network for Graph-based Fraud Detection with Extremely Limited Labels (CGNN)

2025 | AAAI | Extremely limited labels | Reproducibility: MEDIUM

Problem Statement / Simplified Abstract Explicitly targets graph fraud detection with very few labelled examples using contextual decomposition, denoising attention, augmentation and consistency learning.

Core & Fraud Relevance A recent reference for how severe label scarcity should be tested. Code is reported by specialist anomaly-detection lists, but not consistently surfaced by the main fraud-paper repo.

Project Fit / Research Gap RQ2. Experiment value: stress-test Tier-1 models at very low labelled fractions; implement CGNN only if code feasibility is confirmed.

Limitations & Future Work No author-stated future-work claim is treated as established here unless explicitly verified in the primary source. For Tier 2, the project value is primarily experimental: use the paper to define a comparator, ablation or stress condition under the common protocol.

Reproducibility Decision MEDIUM priority relative to Tier 1. Promote to implementation only after the early feasibility gate confirms repository health, dataset access, framework compatibility, licence and memory requirements.

12. Mitigating Message Imbalance in Fraud Detection with Dual-View Graph Representation Learning (MimbFD)

2025 | IJCAI | Message imbalance | Reproducibility: LOW-MEDIUM

Problem Statement / Simplified Abstract Shows that imbalance can occur inside message passing even when the loss is balanced: benign messages can overwhelm rare fraud supervision. It adds topological reachability and local debiasing.

Core & Fraud Relevance Very important conceptual comparator to PC-GNN; implementation priority is lower because official author code is not clearly exposed by the main repository.

Project Fit / Research Gap RQ2/RQ4. Experiment value: add message/neighbour composition statistics to explain why PC-GNN/PMP succeed or fail.

Limitations & Future Work No author-stated future-work claim is treated as established here unless explicitly verified in the primary source. For Tier 2, the project value is primarily experimental: use the paper to define a comparator, ablation or stress condition under the common protocol.

Reproducibility Decision MEDIUM priority relative to Tier 1. Promote to implementation only after the early feasibility gate confirms repository health, dataset access, framework compatibility, licence and memory requirements.

13. Dynamic Neighborhood Modeling via Node-Subgraph Contrastive Learning for Graph-Based Fraud Detection (DCL-GFD)

2025 | AAAI | Dynamic neighbourhoods & contrastive learning | Reproducibility: MEDIUM

Problem Statement / Simplified Abstract Combines node-subgraph contrastive learning with dynamic neighbourhood handling for mixed homophily/heterophily and rare fraud labels.

Core & Fraud Relevance A strong current structural comparator. Use if an implementation becomes reliably available; otherwise it still informs stress-test design.

Project Fit / Research Gap RQ2/RQ4. Experiment value: compare dynamic neighbourhood logic with fixed GraphSAGE and partitioned PMP.

Limitations & Future Work No author-stated future-work claim is treated as established here unless explicitly verified in the primary source. For Tier 2, the project value is primarily experimental: use the paper to define a comparator, ablation or stress condition under the common protocol.

Reproducibility Decision MEDIUM priority relative to Tier 1. Promote to implementation only after the early feasibility gate confirms repository health, dataset access, framework compatibility, licence and memory requirements.

14. TRAVELFRAUDBENCH: A Configurable Evaluation Framework for GNN Fraud Ring Detection in Travel Networks

2026 | arXiv preprint | Benchmark leakage & topology-controlled stress tests | Reproducibility: HIGH

Problem Statement / Simplified Abstract A configurable open-source benchmark with ring-based splits, multiple fraud-ring topologies and explicit topology ablations. Its domain is synthetic travel fraud, but the split design directly addresses leakage and structural stress testing.

Core & Fraud Relevance MIT-licensed Python package with PyG/DGL/NetworkX exporters and public generated datasets. Best used as benchmark-design inspiration, not a replacement for core fraud datasets.

Project Fit / Research Gap RQ4/reproducibility. Experiment value: keep connected fraud rings entirely within one partition when relevant, and borrow topology-controlled ablation ideas.

Limitations & Future Work No author-stated future-work claim is treated as established here unless explicitly verified in the primary source. For Tier 2, the project value is primarily experimental: use the paper to define a comparator, ablation or stress condition under the common protocol.

Reproducibility Decision HIGH priority relative to Tier 1. Promote to implementation only after the early feasibility gate confirms repository health, dataset access, framework compatibility, licence and memory requirements.

15. Alleviating the Inconsistency Problem of Applying Graph Neural Network to Fraud Detection (GraphConsis)

2020 | SIGIR | Graph inconsistency & relation-aware aggregation | Reproducibility: HIGH

Problem Statement / Simplified Abstract A foundational pre-CARE fraud paper that diagnoses inconsistency between node features, labels and neighbourhood information and develops a relation-aware aggregation strategy. It is especially useful for explaining why ordinary GCN-style aggregation can fail on fraud graphs.

Core & Fraud Relevance The safe-graph index exposes both the paper and code. It is older than CARE-GNN but historically important and directly compatible with the projects RQ4 story about neighbourhood composition and graph characteristics.

Project Fit / Research Gap RQ1/RQ4. Experiment value: use mainly as a historical coded comparator or an ablation reference showing the progression GraphConsis -> CARE-GNN -> PMP rather than implementing every generation.

Limitations & Future Work No author-stated future-work claim is treated as established here unless explicitly verified in the primary source. For Tier 2, the project value is primarily experimental: use the paper to define a comparator, ablation or stress condition under the common protocol.

Reproducibility Decision HIGH priority relative to Tier 1. Promote to implementation only after the early feasibility gate confirms repository health, dataset access, framework compatibility, licence and memory requirements.

16. GAGA: Label Information Enhanced Fraud Detection against Low Homophily in Graphs

2023 | The Web Conference | Low homophily & label-enhanced aggregation | Reproducibility: HIGH

Problem Statement / Simplified Abstract GAGA targets low-homophily fraud graphs by injecting label/group information into aggregation and Transformer-style modelling, explicitly using the small amount of labelled data to preserve fraud-relevant signals.

Core & Fraud Relevance Official paper and code are surfaced by the safe-graph index. It directly bridges RQ2 label scarcity with RQ4 homophily and therefore is a stronger experiment-design paper than several generic anomaly methods.

Project Fit / Research Gap RQ2/RQ4. Experiment value: compare whether label-enhanced grouping remains useful as labelled positives are reduced and whether its advantage tracks measured fraud-node homophily.

Limitations & Future Work No author-stated future-work claim is treated as established here unless explicitly verified in the primary source. For Tier 2, the project value is primarily experimental: use the paper to define a comparator, ablation or stress condition under the common protocol.

Reproducibility Decision HIGH priority relative to Tier 1. Promote to implementation only after the early feasibility gate confirms repository health, dataset access, framework compatibility, licence and memory requirements.

17. LEX-GNN: Label-Exploring Graph Neural Network for Accurate Fraud Detection

2024 | CIKM | Label exploration & scarce supervision | Reproducibility: HIGH

Problem Statement / Simplified Abstract LEX-GNN is a fraud-specific label-exploration method designed to make better use of limited labelled information instead of treating the unknown pool as ordinary benign data.

Core & Fraud Relevance The safe-graph list provides both paper and code links. It is a strong recent reference for RQ2, but should be promoted to implementation only if its repository adapts cleanly to YelpChi/Amazon and the common split registry.

Project Fit / Research Gap RQ2. Experiment value: label-budget sweep with fixed test distribution; compare label-exploration against PC-GNN sampling and a weighted-loss GraphSAGE baseline.

Limitations & Future Work No author-stated future-work claim is treated as established here unless explicitly verified in the primary source. For Tier 2, the project value is primarily experimental: use the paper to define a comparator, ablation or stress condition under the common protocol.

Reproducibility Decision HIGH priority relative to Tier 1. Promote to implementation only after the early feasibility gate confirms repository health, dataset access, framework compatibility, licence and memory requirements.

18. Global Attribute-Association Pattern Aggregation for Graph Fraud Detection (GAAP)

2025 | AAAI | Global attribute-association patterns | Reproducibility: HIGH

Problem Statement / Simplified Abstract GAAP addresses the fact that fraud can be expressed through global attribute-association patterns that local neighbourhood aggregation or crude feature binning may miss. It learns and globally aggregates attribute-association patterns to distinguish fraud from benign behaviour.

Core & Fraud Relevance AAAI provides the primary paper and the safe-graph list exposes an author-code link. It adds a genuinely different feature/global-pattern family without changing the supervised node-classification task.

Project Fit / Research Gap RQ1/RQ4, secondary RQ3. Experiment value: use as a modern coded reserve to test whether global association modelling adds enough PR-AUC to justify its complexity over local message-passing baselines.

Limitations & Future Work No author-stated future-work claim is treated as established here unless explicitly verified in the primary source. For Tier 2, the project value is primarily experimental: use the paper to define a comparator, ablation or stress condition under the common protocol.

Reproducibility Decision HIGH priority relative to Tier 1. Promote to implementation only after the early feasibility gate confirms repository health, dataset access, framework compatibility, licence and memory requirements.

Tier 3 - Literature Review, Boundaries and Future Directions

Tier 3 prevents the literature review from becoming narrow while protecting the implementation from scope creep. These papers are useful for historical progression, deployment context, explainability, cold-start, temporal or LLM directions, but they are not the best use of the reproducibility budget for the current proposal.

1. FRAUDRE: Fraud Detection Dual-Resistant to Graph Inconsistency and Imbalance

2021 | ICDM | Imbalance + inconsistency

Simplified Abstract / Relevance Important historical fraud-specific method, but weaker reproducibility/implementation fit than PC-GNN and newer methods. Cite as a bridge between imbalance and graph inconsistency.

Project Fit LITERATURE / FUTURE-WORK SUPPORT. Keep as a citation or discussion boundary; do not spend core implementation time unless the supervisor changes scope or a Tier-1 method fails feasibility.

2. MAFI: Multiple Aggregators and Feature Interactions Network for Fraud Detection over Heterogeneous Graph

2021 | IEEE Transactions on Big Data | Heterogeneous aggregation

Simplified Abstract / Relevance Useful for heterogeneous-graph literature and complexity discussion, but less aligned with the guaranteed static YelpChi/Amazon benchmark and not necessary if PMP/DiG-In-GNN already cover relation-aware modelling.

Project Fit LITERATURE / FUTURE-WORK SUPPORT. Keep as a citation or discussion boundary; do not spend core implementation time unless the supervisor changes scope or a Tier-1 method fails feasibility.

3. AUC-oriented Graph Neural Network for Fraud Detection

2022 | The Web Conference | Imbalance-aware optimisation

Simplified Abstract / Relevance A useful example of changing the training objective rather than graph sampling. Keep for literature/metric discussion; implementing it adds less coverage than PC-GNN because PR-AUC remains the projects primary evaluation metric.

Project Fit LITERATURE / FUTURE-WORK SUPPORT. Keep as a citation or discussion boundary; do not spend core implementation time unless the supervisor changes scope or a Tier-1 method fails feasibility.

4. Dual-Augment Graph Neural Network for Fraud Detection

2022 | CIKM | Graph augmentation & camouflage

Simplified Abstract / Relevance Relevant structural augmentation work, but author-code feasibility is weaker than current alternatives. Cite when discussing graph modification as a response to camouflage.

Project Fit LITERATURE / FUTURE-WORK SUPPORT. Keep as a citation or discussion boundary; do not spend core implementation time unless the supervisor changes scope or a Tier-1 method fails feasibility.

5. H2-FDetector: A GNN-based Fraud Detector with Homophilic and Heterophilic Connections

2022 | The Web Conference | Mixed homophily/heterophily

Simplified Abstract / Relevance Important in the historical progression toward PMP/GHRN/SplitGNN. Tier 3 because newer reproducible methods represent the same problem more directly.

Project Fit LITERATURE / FUTURE-WORK SUPPORT. Keep as a citation or discussion boundary; do not spend core implementation time unless the supervisor changes scope or a Tier-1 method fails feasibility.

6. Bi-Level Selection via Meta Gradient for Graph-Based Fraud Detection

2022 | DASFAA | Noisy training information

Simplified Abstract / Relevance Useful for robustness-to-noisy-label discussion, but label-noise is not a guaranteed RQ and no strong code advantage was verified. Keep for literature and optional future work.

Project Fit LITERATURE / FUTURE-WORK SUPPORT. Keep as a citation or discussion boundary; do not spend core implementation time unless the supervisor changes scope or a Tier-1 method fails feasibility.

7. Inductive Graph Representation Learning for Fraud Detection

2022 | Expert Systems with Applications | Unseen-node generalisation

Simplified Abstract / Relevance Operationally relevant, but the current project is explicitly static anomalous-node classification with persistent splits. Use to acknowledge that inductive deployment is outside current scope.

Project Fit LITERATURE / FUTURE-WORK SUPPORT. Keep as a citation or discussion boundary; do not spend core implementation time unless the supervisor changes scope or a Tier-1 method fails feasibility.

8. xFraud: Explainable Fraud Transaction Detection

2022 | PVLDB | Production scale & explainability

Simplified Abstract / Relevance Excellent real-world evidence at eBay scale and extremely low fraud prevalence, but proprietary data makes direct reproduction unrealistic. Cite for why runtime, scalability and explanation matter.

Project Fit LITERATURE / FUTURE-WORK SUPPORT. Keep as a citation or discussion boundary; do not spend core implementation time unless the supervisor changes scope or a Tier-1 method fails feasibility.

9. SEFraud: Graph-based Self-Explainable Fraud Detection via Interpretative Mask Learning

2024 | KDD | Interpretability

Simplified Abstract / Relevance Strong self-explainable fraud work. The project initiation already treats interpretability as secondary/conditional, and public author code is not clearly available; keep as responsible-research context.

Project Fit LITERATURE / FUTURE-WORK SUPPORT. Keep as a citation or discussion boundary; do not spend core implementation time unless the supervisor changes scope or a Tier-1 method fails feasibility.

10. VecAug: Unveiling Camouflaged Frauds with Cohort Augmentation for Enhanced Detection

2024 | KDD | Cohort-based camouflage

Simplified Abstract / Relevance Relevant modern camouflage approach, but it overlaps with the Tier-1/2 heterophily suite and is not needed for a semester-scale reproducible core.

Project Fit LITERATURE / FUTURE-WORK SUPPORT. Keep as a citation or discussion boundary; do not spend core implementation time unless the supervisor changes scope or a Tier-1 method fails feasibility.

11. Alfrad: Early Fraud Detection with Augmented Graph Learning

2020 | DLG @ KDD workshop | Early/cold-start fraud

Simplified Abstract / Relevance A creative model-agnostic augmentation wrapper with a genuine low-data angle. Keep as future extension for early-stage/cold-start analysis, not as core due age/workshop status and different setup.

Project Fit LITERATURE / FUTURE-WORK SUPPORT. Keep as a citation or discussion boundary; do not spend core implementation time unless the supervisor changes scope or a Tier-1 method fails feasibility.

12. DFraud3: Multi-Component Fraud Detection Free of Cold-start

2020 | arXiv | Cold-start & inductive learning

Simplified Abstract / Relevance Uses Yelp-like data and motivates unseen-user evaluation, but the current project intentionally keeps a common static node task. Useful future-work citation.

Project Fit LITERATURE / FUTURE-WORK SUPPORT. Keep as a citation or discussion boundary; do not spend core implementation time unless the supervisor changes scope or a Tier-1 method fails feasibility.

13. MLED: Can LLMs Find Fraudsters? Multi-level LLM Enhanced Graph Fraud Detection

2025 | ACM Multimedia | LLM semantic enhancement

Simplified Abstract / Relevance The previous 2025-26 review treated this as a foundation paper, but the official project initiation contains no LLM research question. It remains valuable frontier literature, not a Tier-1 implementation target.

Project Fit LITERATURE / FUTURE-WORK SUPPORT. Keep as a citation or discussion boundary; do not spend core implementation time unless the supervisor changes scope or a Tier-1 method fails feasibility.

14. FLAG: Fraud Detection with LLM-enhanced Graph Neural Network

2025 | KDD | LLM + semantic neighbour filtering

Simplified Abstract / Relevance Important evidence that LLM semantics can support fraud detection and camouflage handling, but it answers a different project question and adds text/LLM dependencies beyond the reproducibility-first core.

Project Fit LITERATURE / FUTURE-WORK SUPPORT. Keep as a citation or discussion boundary; do not spend core implementation time unless the supervisor changes scope or a Tier-1 method fails feasibility.

15. A Dual-Granularity Prompting Framework for Fraud Detection with Graph-Enhanced LLMs (DGP)

2026 | AAAI | LLM as predictor

Simplified Abstract / Relevance A useful contrast showing a heavier LLM-as-classifier paradigm and prompt-length trade-offs. Literature/future direction only under the current official benchmark scope.

Project Fit LITERATURE / FUTURE-WORK SUPPORT. Keep as a citation or discussion boundary; do not spend core implementation time unless the supervisor changes scope or a Tier-1 method fails feasibility.

16. FraudCoT: Autonomous Chain-of-Thought Distillation for Graph-Based Fraud Detection

2026 | arXiv | LLM reasoning/distillation

Simplified Abstract / Relevance Frontier reasoning work that is compute-heavy and outside the project RQs. Useful for future-work discussion about semantic reasoning versus reproducible graph baselines.

Project Fit LITERATURE / FUTURE-WORK SUPPORT. Keep as a citation or discussion boundary; do not spend core implementation time unless the supervisor changes scope or a Tier-1 method fails feasibility.

17. LGSPF: Let Relations Speak - An End-to-End LLM-GNN Soft Prompt Framework for Fraud Detection

2026 | arXiv | LLM-GNN without text dependency

Simplified Abstract / Relevance Important limitation-driven LLM work for structure-only graphs, but substantially changes the methodology and hardware/software requirements; Tier 3 under the current proposal.

Project Fit LITERATURE / FUTURE-WORK SUPPORT. Keep as a citation or discussion boundary; do not spend core implementation time unless the supervisor changes scope or a Tier-1 method fails feasibility.

18. L2IR: Revealing Latent Intent in Graph Fraud Detection

2026 | arXiv | LLM intent reasoning

Simplified Abstract / Relevance Strong camouflage-focused future direction; not necessary for the current reproducibility benchmark because it introduces LLM inference and per-edge semantic reasoning.

Project Fit LITERATURE / FUTURE-WORK SUPPORT. Keep as a citation or discussion boundary; do not spend core implementation time unless the supervisor changes scope or a Tier-1 method fails feasibility.

19. Enhancing Attribute-Driven Fraud Detection with Risk-Aware Graph Representation (RG TAN)

2025 | IEEE TKDE | Temporal/risk-aware transaction fraud

Simplified Abstract / Relevance A realistic financial-domain method and useful boundary case, but temporal transaction modelling is conditional rather than guaranteed in the initiation methodology.

Project Fit LITERATURE / FUTURE-WORK SUPPORT. Keep as a citation or discussion boundary; do not spend core implementation time unless the supervisor changes scope or a Tier-1 method fails feasibility.

20. PromoGuardian: Detecting Promotion Abuse Fraud with Multi-Relation Fused GNNs

2026 | IEEE S&P / preprint lineage | Production group fraud

Simplified Abstract / Relevance Strong production-scale group-fraud motivation, but the task, data and graph construction differ too much from the guaranteed common node-classification benchmark for direct implementation.

Project Fit LITERATURE / FUTURE-WORK SUPPORT. Keep as a citation or discussion boundary; do not spend core implementation time unless the supervisor changes scope or a Tier-1 method fails feasibility.

21. Nowhere to H2IDE: Fraud Detection From Multi-Relation Graphs via Disentangled Homophily and Heterophily Identification

2024 | IEEE TKDE | Disentangled homophily/heterophily

Simplified Abstract / Relevance A significant multi-relation fraud paper, but the IEEE publication record places it in late 2024 even though the discovery repository groups it under 2025. It has no clear code advantage over PMP/GAGA/CARE-GNN, so retain it as literature context and a conceptual bridge rather than another implementation commitment.

Project Fit LITERATURE / FUTURE-WORK SUPPORT. Keep as a citation or discussion boundary; do not spend core implementation time unless the supervisor changes scope or a Tier-1 method fails feasibility.

22. Grad: Guided Relation Diffusion Generation for Graph Augmentation in Graph Fraud Detection

2025 | The Web Conference | Relation generation & adaptive camouflage

Simplified Abstract / Relevance A strong structural counter-hypothesis: instead of enriching node semantics, Grad repairs weak fraud structure by generating auxiliary relations after contrastive learning. Highly relevant to camouflage, but no code link is surfaced in the main discovery list, so implementation risk is higher than the coded Tier-1/2 alternatives.

Project Fit LITERATURE / FUTURE-WORK SUPPORT. Keep as a citation or discussion boundary; do not spend core implementation time unless the supervisor changes scope or a Tier-1 method fails feasibility.

23. A Label-Free Heterophily-Guided Approach for Unsupervised Graph Fraud Detection (HUGE-GAD)

2025 | AAAI | Unsupervised heterophily-guided fraud detection

Simplified Abstract / Relevance Important because it removes the need for labels and provides code, but the official project is supervised anomalous-node classification with controlled label scarcity rather than zero-label detection. Keep as a boundary citation for the extreme end of RQ2.

Project Fit LITERATURE / FUTURE-WORK SUPPORT. Keep as a citation or discussion boundary; do not spend core implementation time unless the supervisor changes scope or a Tier-1 method fails feasibility.

24. GraphFC: Customs Fraud Detection with Label Scarcity

2023 | CIKM | Label scarcity in customs fraud

Simplified Abstract / Relevance Directly relevant to scarce labels and has a code link, but it changes the domain, graph/data construction and customs-inspection task enough that it is not a clean YelpChi/Amazon comparator. Cite it to show label scarcity is operationally real across fraud domains.

Project Fit LITERATURE / FUTURE-WORK SUPPORT. Keep as a citation or discussion boundary; do not spend core implementation time unless the supervisor changes scope or a Tier-1 method fails feasibility.

Recommended experiment blueprint

Stage 0 - Feasibility gate For every candidate repository, record: official-code status, licence, last-known dependency versions, dataset acquisition path, whether the model can accept the projects persistent node IDs, expected GPU/CPU memory, and estimated adaptation effort. Reject/replace models before large experiments, and log every change from author code.

Stage 1 - Reproduce reference runs Run the smallest author-compatible configuration first. Do not chase paper numbers yet; verify end-to-end loading, training, validation-only model selection, metric calculation and repeatability across two seeds.

Stage 2 - Unified baseline benchmark Run MLP, neighbourhood-aggregated tree ensemble, GCN, GraphSAGE and 2-4 fraud-specific Tier-1 models on YelpChi and Amazon using the exact same split registry, features, validation policy and metric implementation.

Stage 3 - RQ2 stress grid Keep the test set fixed. Reduce labelled training nodes and/or fraud prevalence in a controlled factorial grid. Always retain enough positive examples for meaningful training and report the exact count, not only percentage.

Stage 4 - RQ4 graph-characteristic analysis Compute fraud-node homophily, degree, relation count and neighbourhood label composition. Analyse model deltas by graph-characteristic bins and across datasets. This converts a leaderboard into an explanation of when a model is useful.

Stage 5 - RQ3 resource profiling Measure wall-clock training time, inference time, peak CPU/GPU memory and trainable parameters in one environment. Repeat after warm-up where appropriate; clearly separate data-loading/preprocessing cost from training and inference.

Stage 6 - Optional novelty slice Choose only one: (a) combine label scarcity + imbalance following the gap highlighted by GAD in the Wild; or (b) a small structural/adversarial edge stress following BWGNN/MonTi. A narrow, completed extension is stronger than several half-built architectures.

Paper-to-RQ map for the proposal

Paper	RQ1	RQ2	RQ3	RQ4
GADBench	Strong	Support	Strong	Support
PC-GNN	Strong	Core	Support	Strong
CARE-GNN	Strong	Support	Support	Core
BWGNN	Strong	Strong	Strong	Core
PMP	Strong	Strong	Support	Core
GAD in the Wild	Support	Core	Core	Support
MonTi (optional)	Support	-	Support	Robustness

Paper	RQ1	RQ2	RQ3	RQ4
GHRN / SplitGNN	Strong	Support	Support	Core
Barely Supervised / CGNN	Support	Core	-	Support
FraudGT	Support	-	Core	Support
DGA-GNN / DiG-In-GNN	Strong	Support	Support	Strong
TravelFraudBench	Method	Support	Support	Core
GraphConsis / GAGA	Strong	Support	Support	Core
LEX-GNN	Support	Core	Support	Support
GAAP	Strong	Support	Support	Strong

Reproducibility-first implementation priority

Method / benchmark	Author code	Core-data fit	Integration risk	Recommendation
GADBench	Yes	High	Low-Med	Use framework ideas
PC-GNN	Yes	High	Medium	Implement
CARE-GNN	Yes	High	Medium	Implement if time
BWGNN	Yes	High	Low-Med	Implement
PMP	Yes	High	Medium	Implement
GHRN	Yes	High/Med	Medium	Reserve
SplitGNN	Yes	High/Med	Medium	Reserve
DGA-GNN	Yes	YelpChi + others	Medium	Reserve
DiG-In-GNN	Yes	Multi-relation	Medium	Reserve
SEC-GFD	Yes	High	Medium	Reserve
Barely Supervised	No clear official repo	High conceptually	High	Design reference
FraudGT	Yes	Conditional	Medium-High	Conditional
MonTi	Yes	Attack pipeline	High	Optional extension
GAD in the Wild	Yes (release)	Methodology	Medium	Design reference

Method / benchmark	Author code	Core-data fit	Integration risk	Recommendation
TravelFraudBench	Yes/package	Different domain	Low	Design reference
GraphConsis	Yes	High	Medium	Historical reserve
GAGA	Yes	High	Medium	Reserve
LEX-GNN	Yes	High conceptually	Medium	Reserve
GAAP	Yes	Fraud graphs	Medium	Reserve

Supervisor meeting shortlist

Do not bring the entire review into the discussion as if every paper is an implementation commitment. The meeting should decide the model suite and exactly one novelty/stress extension.

1. **GADBench** - Benchmark foundation: justify why a unified framework is needed and what GADBench still leaves open for a fraud-focused, controlled benchmark.
2. **PC-GNN** - Imbalance foundation: directly named in the project; official code; simple story for RQ2.
3. **PMP** - Best current message-passing foundation for RQ4: mixed homophily/heterophily and imbalance, with public code.
4. **BWGNN** - Strong spectral counter-family and highly reproducible reference; explicit trusted-edge limitation offers a possible structural stress extension.
5. **CARE-GNN** - Historical camouflage foundation and multi-relation baseline on the same fraud datasets.
6. **GAD in the Wild** - Recent benchmark paper that validates controlled realistic stress testing and explicitly leaves combined challenge axes open.
7. **MonTi - optional** - Only bring as an optional novelty question: is a narrow robustness stress test worth the added engineering cost?

Researcher verdict on the supplied drafts

21-22 member draft Useful and readable, with a correct fair-comparison motivation and sensible experiment ideas. Its weakness is prioritisation: it promotes many older models without first checking official code, dataset/task compatibility or whether a newer reproducible paper already represents the same mechanism. Researcher rating: 7.5/10 as a literature draft; about 6.5/10 as an implementation-selection document.

Individual Deep Dives draft Better at distinguishing direct benchmark candidates from proprietary or adjacent-domain papers, and especially good at flagging xFraud as valuable but not directly reproducible. It still lacks a unified RQ-based tier system and does not consistently map author limitations/future work to a feasible project contribution. Researcher rating: 8/10 as a relevance screen.

Previous 2025-26 PDF Strong paper-level writing and research-gap structure, but it incorrectly treated the project as an LLM-enhancement benchmark. Compared with the official initiation document, that is a strategic mismatch. The LLM papers remain valuable Tier-3 frontier literature, not the foundation of the current reproducible benchmark. Researcher rating: 8.5/10 as an LLM review, about 5.5/10 for alignment to the official current project.

Combined review here This version is stronger for the proposal because it starts from the signed-off task/RQs and feasibility gate, then uses recency and novelty as secondary criteria. It deliberately promotes fewer models, makes code/data compatibility visible, and turns limitations/future work into concrete experiment slices rather than a list of architecture ideas.

Verification note

The safe-graph/graph-fraud-detection-papers repository was used as a discovery index across years, but inclusion was not based on the repository title alone. Major selected papers were cross-checked against official ACM, AAAI, ICLR/OpenReview, NeurIPS, PMLR/IJCAI or arXiv records and, where implementation priority mattered, against the author repository. The document intentionally does not reproduce a link dump; direct meeting links are supplied separately in chat.

Terminology note: “Research Gap -> Project Objectives” is a researcher synthesis unless the text explicitly says AUTHOR-STATED. This distinction is important: a paper not mentioning your exact benchmark is not evidence that its authors claimed your research gap.